Pollaris
Software Development Plan (Small Project)
Version 1.0

Project: Pollaris – Community-Based Decision-Making Web Platform

Project Team: Group 05

Team members:

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18 October , 2026

Revision History

Date	Version	Description	Author
18/Oct/2026	1.0	Initial Software Development Plan drafted for Inception Phase (PA1). Defines project purpose, scope, team organization, 5-sprint schedule, test builds, requirements management, and risk matrix.	Group 05

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Software Development Plan (Small Project)

1. Introduction

1.1 Purpose

This Software Development Plan defines the development process, schedule, and team organization for the Pollaris web application. Our team is developing Pollaris as our semester project for the course Introduction to Software Engineering (Class 24C01) at the Faculty of Information Technology, University of Science – VNU-HCM. This plan coordinates our team's work across five two-week sprints and helps our instructors and teaching assistants evaluate our progress throughout the semester.

1.2 Scope

This plan covers the engineering lifecycle of Pollaris from initial project setup in October 2026 to final release in December 2026. It governs user research, system architecture, frontend and backend implementation, database setup, AI integration, and testing. Detailed feature requirements and system specifications are documented in the companion Pollaris Vision Document. Our team tracks individual tasks on our Jira Scrum board.

1.3 Overview

The remainder of this document is organized into three main sections. Section 2 summarizes the project purpose, target users, assumptions, constraints, and course deliverables. Section 3 describes our team structure and member responsibilities. Section 4 presents the phase plan, iteration objectives, release schedule, Jira work breakdown, resourcing, requirements management, and risk management.

2. Project Overview

2.1 Project Purpose, Scope, and Objectives

People often face difficulty making everyday decisions, such as choosing tech gadgets, planning group trips, or picking restaurants. Asking for advice on social media usually leads to cluttered comment sections that are hard to compare. Pollaris is a responsive web application where users create polls, vote, discuss options, and view AI-generated summaries of community opinions and pros and cons.

Target users include students, online shoppers, social media users, and friend groups. The system scope covers **7 core MVP feature groups (M01 to M07)** as defined in the Pollaris Vision Document: User Identity & Access Management (M01), Poll Creation & Management (M02), Voting & Decision Results (M03), Community Discussion & Feedback (M04), AI-Powered Opinion Analysis & Personalized Decision Support (M05), Poll Discovery & Sharing (M06), and Community Safety & Moderation (M07). For the complete description of each feature group, please refer to the companion Pollaris Vision Document.

Our main objectives are to deliver a working web application within 10 weeks, ensure voting is reliable and consistent, and generate accurate AI summaries from user comments.

2.2 Assumptions and Constraints

Our plan is based on the following assumptions and constraints:

- Users access Pollaris through modern web browsers with an active internet connection.

- An external LLM API service remains accessible for generating opinion summaries.
- Each team member contributes approximately 10 to 15 hours per week.
- The project must be completed within 10 weeks across five two-week sprints, ending on December 13, 2026.
- The team operates with zero budget, using free-tier developer tools and hosting.
- The team consists of four undergraduate students who share development and testing duties.

2.3 Project Deliverables

The project produces deliverables aligned with the course schedule detailed in PA0:

Table 2.1: Project Deliverables and Target Delivery Schedule

Phase (RUP)	Sprint	Milestone	Due Date	Deliverables	Responsible
Inception	1	PA0 & PA1	18/Oct/2026	Project Proposal, Team Registration, Tools Setup (Slack, Jira, GitHub), Project Plan v1.0, Vision Document v1.0, 3–6 UI sketches, Weekly Reports 1 & 2.	All Members
Elaboration	2	PA2	01/Nov/2026	Requirements analysis, Use Case specifications, initial software architecture, UI wireframes, Weekly Reports 3 & 4.	All Members
Elaboration	3	PA3	15/Nov/2026	Detailed design (UML diagrams, database schema), Test plan, Working Software Build 1, Weekly Reports 5 & 6.	All Members
Construction	4	PA4	29/Nov/2026	Working Software Build 2, integration testing, Weekly Reports 7 & 8.	All Members
Construction	5	PA5	13/Dec/2026	Final Working Software Build 3, final testing, user guide, demonstration video, Weekly Reports 9 & 10.	All Members

3. Project Organization

3.1 Organizational Structure

Our team structure connects academic supervision with an agile Scrum development workflow:

- Course Review Authorities: Teacher Nguyễn Văn Vũ, Teacher Trương Phước Lộc, Teacher Hồ Tuấn Thanh, and course Teaching Assistants. They review deliverables, evaluate milestones, and provide feedback after each assignment.
- Team Group 05: A team of four students. Nguyễn Thành Tuân acts as Project Manager and Scrum Master. All four members participate in design, coding, code reviews, and testing to maintain a balanced workload.

3.2 Roles and Responsibilities

Responsibilities are balanced across all four members according to their roles and tasks:

Table 3.1: Project Team Roles and Responsibilities Matrix

Team Member & ID	Project Role	Key Responsibilities

Team Member & ID	Project Role	Key Responsibilities
Teacher Nguyễn Văn Vũ, Teacher Trương Phước Lộc, Teacher Hồ Tuấn Thanh, and TAs	Course Instructors & TAs	Evaluate project deliverables, review team progress on Jira and GitHub, provide sprint feedback, and grade assignments.
Nguyễn Thành Tuân (24127579)	Project Manager & Full-Stack Developer	Maintain project schedule and Jira sprint board, organize team meetings, compile weekly reports, package assignment submissions, and develop user authentication and poll creation features.
Mạc Văn Hải An (24127316)	Business Analyst & Developer	Lead user survey analysis, write and maintain the Vision Document and Use Case specifications, develop commenting and discussion thread features, and assist in test case execution.
Trần Hữu Phước (24127511)	UI/UX Designer & Frontend Developer	Design UI sketches and wireframes, build responsive web pages, create chart components for poll statistics, develop search and bookmarking interfaces, and conduct UI testing.
Dư Bá Khoa (24127059)	DevOps, Tool Master & Backend Developer	Set up and maintain Slack, Jira, and GitHub repositories, design the database schema, implement the voting engine and AI summarization service, and lead automated test execution.

4. Management Process

4.1 Project Estimates

4.2 Project Plan

4.2.1 Phase Plan

The project follows the Rational Unified Process (RUP) phases across five two-week Scrum sprints:

- Inception (Sprint 1: 05/10/2026 – 18/10/2026): Problem scope, user survey, tool setup, initial plan, vision document, and UI sketches. Major milestone: Lifecycle Objectives (LCO).
- Elaboration (Sprint 2 & 3: 19/10/2026 – 15/11/2026): Detailed requirements, use case specifications, architecture baseline, database design, test plan, and Working Software Build 1. Major milestone: Lifecycle Architecture (LCA).
- Construction (Sprint 4 & 5: 16/11/2026 – 13/12/2026): Implementation of comments, statistics, AI opinion summarization, decision assistant, search, notifications, moderation, and Working Software Build 2 and Build 3. Major milestone: Initial Operational Capability (IOC).

Table 4.1: Project Phase Plan, Sprints, and Milestones

RUP Phase	Sprint	Milestone	Target Date	Key Focus & Deliverables
Inception	1	**LCO**	18/Oct/2026	Feasibility, user survey, tools setup, Project Plan v1.0, Vision Document v1.0, UI sketches.
Elaboration	2	Architecture Baseline	01/Nov/2026	Use Case specs, Domain Model, Software Architecture Document draft, UI wireframes.
Elaboration	3	**LCA**	15/Nov/2026	Detailed UML design, PostgreSQL schema, Test Plan, Working Software Build 1.
Construction	4	Feature Delivery	29/Nov/2026	Comments system, poll statistics charts, AI summarization pipeline, Working Software Build 2.
Construction	5	**IOC**	13/Dec/2026	AI decision assistant, search, notifications, moderation, cloud deploy, Working Software Build 3.

4.2.2 Iteration Objectives

Our team has defined objectives for each sprint:

- Sprint 1 (Inception, 05/10 – 18/10): Register team Group 05, conduct user survey, set up Slack, Jira, and GitHub, author Project Plan and Vision Document, design 3–6 UI sketches, submit Weekly Reports 1 & 2.
- Sprint 2 (Elaboration 1, 19/10 – 01/11): Refine requirements from TA feedback, author detailed Use Case specifications, build Domain Model, draft Software Architecture Document, create interactive wireframes.
- Sprint 3 (Elaboration 2, 02/11 – 15/11): Design UML class and sequence diagrams, create PostgreSQL schema, author Test Plan, implement and test Working Software Build 1 (auth, poll creation, basic voting).
- Sprint 4 (Construction 1, 16/11 – 29/11): Implement comments with option tags, dynamic poll statistics charts, AI opinion summary integration, test Working Software Build 2.
- Sprint 5 (Construction 2, 30/11 – 13/12): Implement AI decision assistant, search, saved polls, notifications, moderation dashboard, deploy Working Software Build 3, write user guide, record demo video.

4.2.3 Releases and Test Builds

We will produce three working software builds for testing and evaluation during the semester. Build 3 is scheduled for December 6, 2026, leaving one week of buffer before the final course deadline on December 13, 2026, for testing, bug fixing, and documentation:

Table 4.2: Summary of Software Releases and Test Builds

Release / Build	Target Date	Intended Audience	Core Scope & Features Included	Verification Goals

Release / Build	Target Date	Intended Audience	Core Scope & Features Included	Verification Goals
Build 1 (Internal Alpha)	15/Nov/2026	Team & TAs	User authentication, database migrations, poll creation, and basic voting engine.	Verify client-server connectivity, test database operations, ensure single-choice voting works.
Build 2 (Beta Build)	29/Nov/2026	Team & TAs	Threaded comments, dynamic poll statistics charts, and AI opinion summary for polls with comments.	Integration testing between frontend and backend; verify AI summary generation and error handling.
Build 3 (Final Release)	06/Dec/2026	Public Users, Instructors & TAs	Full integration of all 7 MVP feature groups (M01–M07): completing AI decision assistance (M05), poll discovery and sharing (M06), and moderation dashboard (M07), fully integrated with core voting and discussion modules.	End-to-end testing, responsive design verification across devices, zero high-severity defects.

4.2.4 Project Schedule

Our project schedule is tracked on Jira under project Group05-Pollaris. All 7 MVP feature groups are broken down into concrete development and testing tasks, with responsibilities balanced across all four members:

Table 4.3: Work Breakdown Structure (WBS) and Jira Sprint Schedule

Task ID	Task Summary & Description	Sprint	Assignee	Start Date	Due Date	Jira Status
POL-1	[PA0] Group Registration & Proposal Coordination: Register Group 05, coordinate brainstorming, consolidate proposal.	Sprint 1	Tuân	05/10/2026	11/10/2026	**DONE**
POL-2	[PA0] User Survey & Problem Scope: Conduct survey with potential users, record notes, identify pain points, define features.	Sprint 1	An	05/10/2026	11/10/2026	**DONE**
POL-3	[PA0] Target Users & Environment Analysis: Define personas, operating environments (Mobile/Web, OS), sketch concept flow.	Sprint 1	Phước	05/10/2026	11/10/2026	**DONE**
POL-4	[PA0] Team Tools Setup & Repository Init: Setup Slack, initialize GitHub repository (/src, /docs, /pa), create Jira board.	Sprint 1	Khoa	05/10/2026	11/10/2026	**DONE**
POL-5	[PA1] Draft Project Plan Document:	Sprint 1	Tuân	12/10/2026	17/10/2026	**IN**

Task ID	Task Summary & Description	Sprint	Assignee	Start Date	Due Date	Jira Status
	Author Software Development Plan (Intro, Overview, Organization, Schedule, Risks).					**PROGRESS**
POL-6	[PA1] Compile Weekly Reports & Final Package: Consolidate Weekly Reports 1 & 2 with Jira screenshots, package zip.	Sprint 1	Tuân	14/10/2026	18/10/2026	**IN PROGRESS**
POL-7	[PA1] Write Vision Document: Complete Vision Document: Problem Statement, User Personas, Features, and NFRs.	Sprint 1	An	12/10/2026	17/10/2026	**IN PROGRESS**
POL-8	[PA1] Design 3-6 App Prototype Screens: Create 3–6 core UI mockups (Auth, Home, Poll Creation, Voting, Results, Summary).	Sprint 1	Phước	12/10/2026	17/10/2026	**IN PROGRESS**
POL-9	[PA1] Jira Sprint Tracking & Evidence Export: Maintain Sprint 1 board, ensure fields are complete, export screenshots.	Sprint 1	Khoa	12/10/2026	17/10/2026	**IN PROGRESS**
POL-10	[PA1] Git Deliverables Sync & Collaboration Logs: Sync deliverables to GitHub /docs and /pa/pa1, collect commit logs.	Sprint 1	Khoa	14/10/2026	18/10/2026	**IN PROGRESS**
POL-11	[PA2] Use Case Specifications: Write formal Use Case specs and activity diagrams for Poll Creation, Voting, and Comments.	Sprint 2	An	19/10/2026	26/10/2026	TO DO
POL-12	[PA2] Software Architecture Document Draft: Design layered architecture, REST API contracts, and database models.	Sprint 2	Khoa	22/10/2026	30/10/2026	TO DO
POL-13	[PA2] High-Fidelity UI Wireframes: Construct interactive Figma prototype for primary user flows.	Sprint 2	Phước	20/10/2026	29/10/2026	TO DO
POL-14	[PA2] Weekly Reports 3 & 4 Compilation: Track sprint progress and compile Weekly Reports 3 & 4.	Sprint 2	Tuân	26/10/2026	01/11/2026	TO DO
POL-15	[PA3] Database Schema & Data Models: Design PostgreSQL schema with constraints, foreign keys, and migrations.	Sprint 3	Khoa	02/11/2026	08/11/2026	TO DO

Task ID	Task Summary & Description	Sprint	Assignee	Start Date	Due Date	Jira Status
POL-16	[PA3] Test Plan Authoring & Test Cases: Define testing strategy and author test cases for auth, polls, and voting.	Sprint 3	An	04/11/2026	11/11/2026	TO DO
POL-17	[PA3] Authentication & User Profile Feature: Implement user registration, login, session tokens, and profile editing.	Sprint 3	Tuân	02/11/2026	12/11/2026	TO DO
POL-18	[PA3] Poll Creation & Core Voting Engine: Implement poll authoring form, options list, and atomic voting mechanism.	Sprint 3	Phước	02/11/2026	13/11/2026	TO DO
POL-19	[PA3] Build 1 Testing & Verification: Execute integration tests for Build 1, verify voting constraints.	Sprint 3	Khoa	12/11/2026	15/11/2026	TO DO
POL-20	[PA4] Comments & Discussion Threads: Implement threaded replies, option voter badges, and comment upvotes.	Sprint 4	An	16/11/2026	22/11/2026	TO DO
POL-21	[PA4] Dynamic Poll Statistics Charts: Build interactive bar and pie charts for real-time percentage visualization.	Sprint 4	Phước	18/11/2026	25/11/2026	TO DO
POL-22	[PA4] AI Opinion Summary Integration: Implement prompt template, external LLM API call, and response parsing.	Sprint 4	Khoa	16/11/2026	26/11/2026	TO DO
POL-23	[PA4] Build 2 Testing & Bug Fixing: Test Build 2 components, verify chart rendering and AI summary outputs.	Sprint 4	Tuân	24/11/2026	29/11/2026	TO DO
POL-24	[PA5] AI Decision Assistant Feature: Implement decision recommendation based on user criteria and community feedback.	Sprint 5	Tuân	30/11/2026	04/12/2026	TO DO
POL-25	[PA5] Search, Filtering & Saved Polls: Build keyword search, category filters, and bookmarking collection.	Sprint 5	Phước	30/11/2026	05/12/2026	TO DO
POL-26	[PA5] Result Sharing & In-App Notifications: Create shareable cards, links, and user notification alerts.	Sprint 5	An	01/12/2026	05/12/2026	TO DO

Task ID	Task Summary & Description	Sprint	Assignee	Start Date	Due Date	Jira Status
POL-27	[PA5] Report & Content Moderation System: Build user report button and administrator moderation panel.	Sprint 5	Khoa	01/12/2026	05/12/2026	TO DO
POL-28	[PA5] Cloud Deployment & Final System Testing: Deploy Build 3 to cloud hosting and verify system end-to-end.	Sprint 5	Khoa & Tuân	06/12/2026	09/12/2026	TO DO
POL-29	[PA5] User Guide & Demo Video: Write user guide manual and record project demonstration video.	Sprint 5	All Members	09/12/2026	13/12/2026	TO DO

4.2.5 Project Resourcing

The team uses the following selected tools and resources throughout the project:

Table 4.4: Project Resourcing, Environment, and Tooling

Resource Category	Selected Tool / Technology	Purpose in Project	Planned Period
Project Management	Jira Software (Scrum)	Sprint backlog, task tracking, and milestone evidence	October – December 2026
Team Communication	Slack & Facebook Group	Team discussions, meeting coordination, and course notifications	October – December 2026
Version Control	GitHub Team Repository	Source code and documentation tracking (/src, /docs, /pa)	October – December 2026
UI/UX Design	AI	Interface sketches, wireframes, and design system	October – November 2026
Development Environment	VS Code & Web Browsers	Code editing, local debugging, and responsive testing	October – December 2026
Web Technologies	Modern Web Frontend & Backend	Responsive web application and RESTful API services	October – December 2026
Database System	Relational Database	Data persistence, transaction management, and vote counting	November – December 2026
AI Service	External LLM API	Opinion summarization and pros/cons extraction	November – December 2026
Cloud Hosting	Free-tier Cloud Hosting	Staging and production deployment for demonstration	November – December 2026

4.3 Project Monitoring and Control

4.3.1 Requirements Management

Requirements for Pollaris are documented in the Vision Document and organized into user stories in our Jira product backlog. If requirements need to be adjusted during development, the team discusses the proposed changes during

sprint planning. Any agreed changes are updated in the Vision Document and logged on the Jira board with revised tasks and assignees.

4.3.2 Reporting and Measurement

4.3.3 Risk Management

We identify and review risks at the start of each sprint. Table 4.5 lists our current technical and non-technical project risks, along with mitigation strategies, owners, and status:

Table 4.5: Project Risk Management Matrix

Ranking	Risk Description and Impact	Mitigation Strategy and Contingency	Owner	Status
High	AI summary hallucination or poor output when comments are few: If the LLM generates arguments not found in comments, users will not trust the summary.	Require at least 5 comments before allowing summary generation. Use strict prompt instructions commanding the model to use only provided comments. Check output against a defined JSON format.	Khoa	**Open**
High	LLM API rate limits, network latency, or free quota depletion: The external AI service may fail or respond slowly during a presentation or testing.	Cache generated summaries by poll ID and comment count so repeat requests return without calling the API again. Display a loading indicator to users. Keep a backup API key ready.	Tuân	**Open**
Medium	Team member unavailability during midterm exams and coursework deadlines: Heavy academic workloads in November may slow down development.	Schedule implementation tasks evenly across sprints. Hold short weekly check-ins to rebalance tasks early if someone is busy with exams.	Tuân	**Open**
Medium	Uneven workload distribution and scheduling conflicts: Some members may carry too many tasks while others have too few, leading to bottlenecks.	Review Jira task assignments weekly. Ensure all four members participate in development and testing tasks.	An	**Open**
Medium	Scope creep across MVP features within the 10-week schedule: Building too many secondary capabilities at once could leave the core voting and AI analysis system unpolished.	Prioritize authentication, poll creation, voting, comments, and AI summaries for Sprints 3 and 4. Keep secondary features modular for Sprint 5.	Tuân	**Open**
Medium	Voting concurrency race conditions and duplicate votes: Concurrent voting requests on the same poll could cause duplicate votes or mismatched counts.	Enforce a unique database constraint on (user_id, poll_id) so a user can vote at most once. Use atomic increment operations in database transactions. Disable the vote button immediately after clicking.	Phước	**Open**
Medium	Accidental leakage of API keys or	Use environment variables (.env) for all	Khoa	**Open**

Ranking	Risk Description and Impact	Mitigation Strategy and Contingency	Owner	Status
	database credentials in public repository: Exposing secrets on GitHub could lead to unauthorized API usage or quota exhaustion.	secrets. Add .env to .gitignore before the first commit. Perform code reviews before merging into develop.		
Low	Free-tier hosting cold starts or downtime during presentation: Free-tier servers may take 30-50 seconds to wake up after inactivity.	Ping the deployed application 10 minutes before demonstrations to wake up the server. Test locally as a fallback during live evaluation.	Khoa	**Open**

4.3.4 Configuration Management